

四川大学期末考试试题（闭卷）

（2023~2024 学年第 1 学期）

A 卷

课程号-课序号: 311232030 / 01 课程名称: 软件工程导论 任课教师: 蒲蔚

适用专业年级: 软件工程 2022 级 学生人数: 22 印题份数: 25 学号: _____ 姓名: _____

考生承诺

我已认真阅读并知晓《四川大学考场规则》和《四川大学本科学生考试违纪作弊处分规定（修订）》，郑重承诺：

1. 已按要求将考试禁止携带的文具用品或与考试有关的物品放置在指定地点；
2. 不带手机进入考场；
3. 考试期间遵守以上两项规定，若有违规行为，同意按照有关条款接受处理。

考生签名: _____

题 号	1 (20%)	2 (20%)	3 (15%)	4 (15%)	5 (20%)	6 (10%)
得 分						
卷面总分			阅卷时间			

注意事项: 1. 请务必将本人所在学院、姓名、学号、任课教师姓名等信息准确填写在答题纸上；

2. 请将答案全部填写在答题纸上；

3. 考试结束，请将试题纸、答卷纸、添卷纸和草稿纸一并交给监考老师。

评阅教师	得分

1. A generic process framework for software engineering encompasses five activities. (20 Points)

(1) What are the five activities?

(2) As a software developer, please outline each activity and its purpose.

注: 试题字迹务必清晰，书写工整。

第1页，共5页
试卷编号: 311-24



扫描全能王 创建

评阅教师	得分

2. Suppose you are developing an ATM for a bank. Material 1 is a software interview record. Material 2 is an observation record of the withdrawal behavior of a bank account. (20 Points)

Material 1:

In an interview to obtain requirements, the customer representative said: "I hope this ATM can support cross bank business. After inserting the card and entering the password, I can choose whether to withdraw or save money. For convenience with withdrawing, some default access amount buttons can be set. I can change my password or report loss. Also, I hope to pay for telephone, water, electricity, and other expenses. For safety reasons, there should be a surveillance camera and a warning bar on the ATM to be cautious of scammers. If the password is entered incorrectly three times, the card should be automatically swallowed."

Material 2:

We obtained the consent of a bank account who agreed to show us his withdrawal process. Here are our observations:
He walked to the ATM machine and looked at the screen (confirming that there were no abnormalities). He inserted the card and immediately looked up. He saw that the ATM asked him to enter the password. He entered the password, clicked on confirm, and then looked up. At this point, the ATM had already switched to the menu interface. At this point, he saw that the "Withdrawal" button was available (he turned his head and said to us, "Sometimes the Withdrawal" button is not available). He clicked on "Withdrawal", entered the withdrawal amount, confirmed, and waited for the ATM machine to count the money, issue it, and remind him to withdraw it. So, he took out the banknote and clicked on "Retrieve Card". The machine successfully exited the card, and he took out the card to complete the withdrawal activity.

- (1) Please analyze the material 1, extract use cases, and establish a use case diagram model.
- (2) Please analyze the material 2 and specify the "withdrawal" use case.
- (3) Please do user interface analysis and design based on the "withdrawal" use case. Please carefully layout each interface, and establish the navigation relationship between the interfaces. (Tip: The withdrawal customer sees the "welcome" screen when coming to the ATM, and then goes to the next screen after he inserts the card.....)



评阅教师	得分

3. Material 3 describes a scenario of how software inner objects collaborate to complete a user request. Please describe this scenario using a sequence diagram and export the class diagram accordingly. (15 Points)

Material 3:

1. The customer enters the request number for the document they want to borrow and clicks the Submit button.
 - 2.1. The Checkout GUI receives the borrowing request, passes the document request number to the CheckController, and delegates(委派) it to handle the borrowing request.
 - 2.2. CheckController first creates an empty msg to provide feedback on the processing results to the customer.
 - 2.3. and then
 - 2.3.1 CheckController uses the document's request number to obtain document d from the database through DBMgr.
 - 2.3.2 DBMgr returns document d to CheckController for judgment. - CheckController determines whether the document exists.
 - 2.3.3 If the document exists (dl!=null)
 - 2.3.3.1 CheckController checks whether the document is available for borrowing
 - 2.3.3.2 If available for borrowing
 - 2.3.3.2.1 CheckController creates a Load object using customer p and document d
 - 2.3.3.2.2 CheckController makes document settings unavailable
 - 2.3.3.2.3 CheckController Using DBMgr to Save Load Objects
 - 2.3.3.2.4 CheckController Saving Documents Using DBMgr
 - 2.3.3.2.5 CheckController writes "checkout successful" to msg
 - 2.3.3.3 Otherwise (not available for borrowing)
 - 2.3.3.3.1 CheckController writes "document not available" to msg
 - 2.3.4 Otherwise (the document does not exist)
 - 2.3.4.1 CheckController writes "document not found" to msg
 - 2.4 CheckController returns msg to Checkout GUI
 - 2.5 Checkout GUI displays msg to customers



评阅教师	得分

4. Material 4 provides a pseudo code that describes the process of printing invoices for each customer. (15Points)

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Material 4:
[Print Invoice]
SalesLedger.open(Sales Ledger 销售分户帐簿)
SalesLedger.getSalesItem
While SalesLedger.hasItems
    Customers.getCustomer(salesItem.customerID)
    Print the customer fetails on the invoice
    While salesItem.customer unchanged
        Print the current salesItem
        Add salesItem.value to total
        SalesLedger.getSaleItem
    EndWhile
    print invoice salesItems total
    Apply customer discounts
    Calculate VAT
    Print amount payable
    Skip to next invoice
End While

```

- (1) Please use a UML activity diagram to describe the process of printing invoices.
- (2) Calculate its loop complexity, where $V(G)=?$
- (3) Design test cases.

注：试题字迹务必清晰，书写工整。

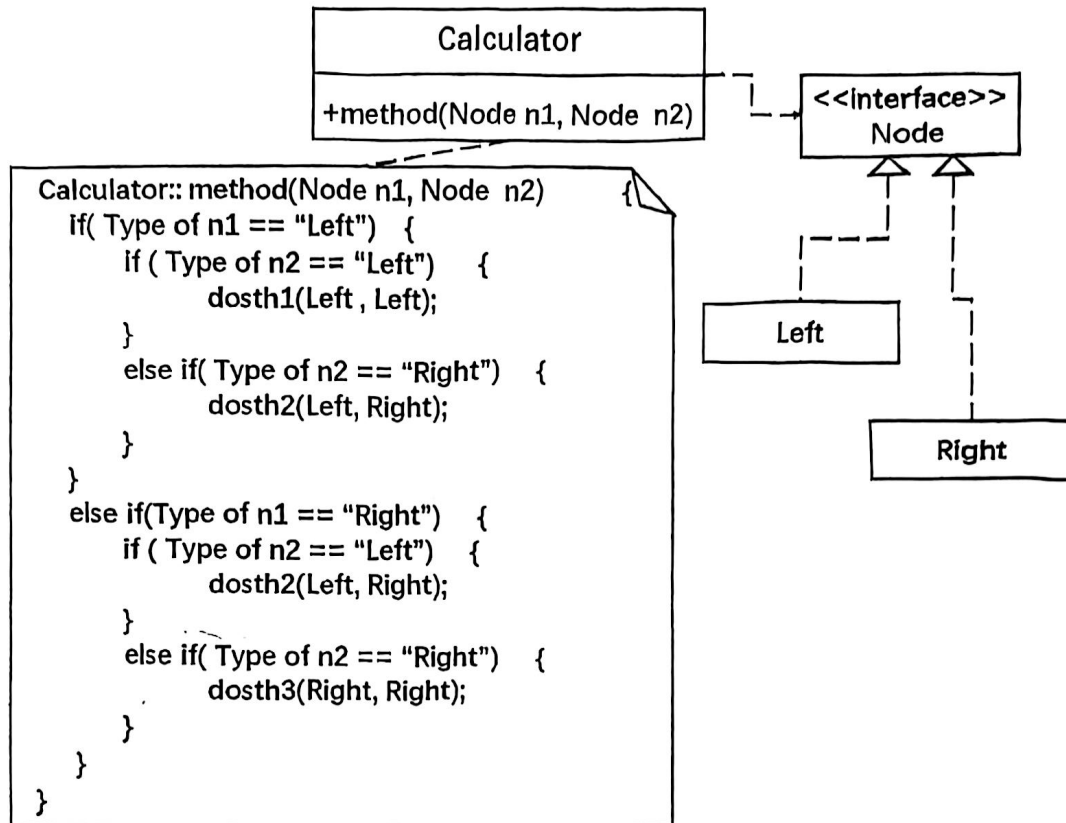
第4页，共5页
试卷编号：311-24



评阅教师	得分

5. Material 5 provides a design solution for a component. (15Points)

Material 5:



- (1) Please explain OCP, LSP, DIP and ISP.
- (2) What are the design issues with it?
- (3) Please refactor and provide a new design solution.

评阅教师	得分

6. Using the topic 'What is the process of software development', use activity diagrams to demonstrate the general process of doing a good job in software development, and explain this process in text. The written explanation shall not be less than 400 words. (10 Points)

注：试题字迹务必清晰，书写工整。



第一题参考答案 (20分)

参考答案

过程框架 (process framework) 定义了若干个框架活动 (framework activity), 为实现完整的软件工程过程建立了基础。这些活动可广泛应用于所有软件开发项目, 无论项目的规模和复杂性如何。此外, 过程框架还包含一些适用于整个软件过程的普适性活动 (umbrella activity)。一个通用的软件工程过程框架通常包含以下5个活动:

五个最基本的过程框架活动是什么?

沟通 在技术工作开始之前, 和客户 (及其他利益相关者^①) 的沟通与协作是极其重要的, 其目的是理解利益相关者的项目目标, 并收集需求以定义软件特性和功能。

策划 如果有地图, 任何复杂的旅程都可以变得简单。软件项目好比是一个复杂的旅程, 策划活动就是创建一个“地图”, 以指导团队的项目旅程, 这个地图称为软件项目计划, 它定义和描述了软件工程师工作, 包括需要执行的技术任务、可能的风险、资源需求、工作产品和工作进度计划。

建模 无论你是庭园设计家、桥梁建造者、航空工程师、木匠还是建筑师, 你每天的工作都离不开模型。你会画一张草图来辅助理解整个项目大的构想——体系结构、不同的构件如何结合, 以及其他一些特征。如果需要, 可以把草图不断细化, 以便更好地理解问题并找到解决方案。软件工程师也是如此, 利用模型来更好地理解软件需求, 并完成符合这些需求的软件设计。

构建 它包括编码 (手写的或者自动生成的) 和测试以发现编码中的错误。

部署 软件 (全部或者部分增量) 交付到用户, 用户对其进行评测并给出反馈意见。

上述五个通用框架活动既适用于简单小程序的开发, 也可用于大型网络应用程序的建造以及基于计算机的大型复杂系统工程。不同的应用案例中, 软件过程的细节可能差别很大, 但是框架活动都是一致的。

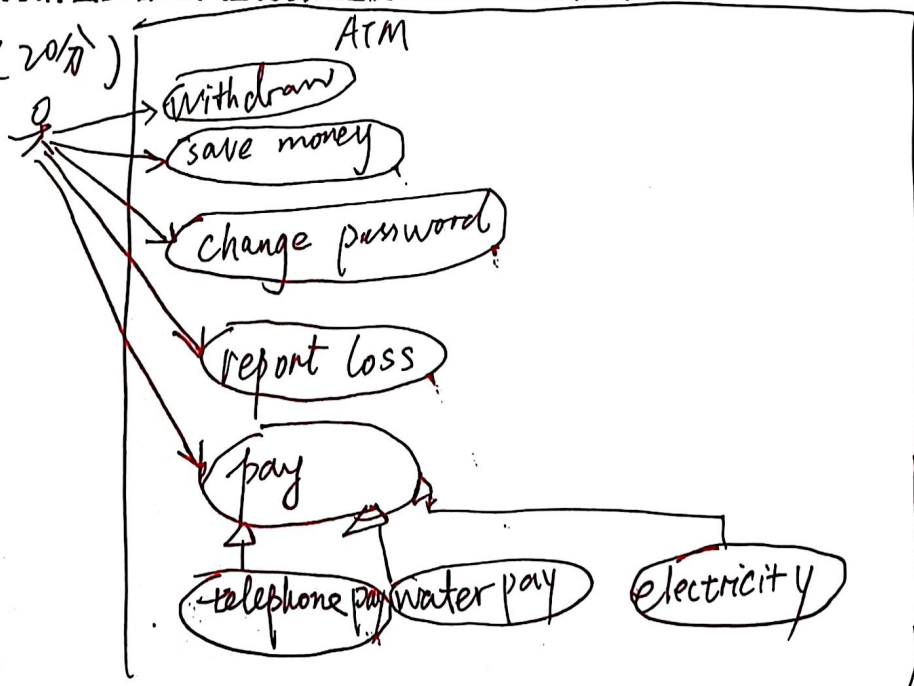
对许多项目来说, 随着项目的开展, 框架活动可以迭代应用。也就是说, 在项目的多次迭代过程中, 沟通、策划、建模、构建、部署等活动不断重复。每次项目迭代都会产生一个软件增量 (software increment), 每个软件增量实现了部分的软件特性和功能。随着每一次增量的产生, 软件逐渐完善。

软件工程过程框架活动由很多普适性活动来补充实现。通常, 这些普适性活动贯穿软件项目始终, 以帮助软件团队管理和控制项目进度、质量、变更和风险。典型的普适性活动包括:

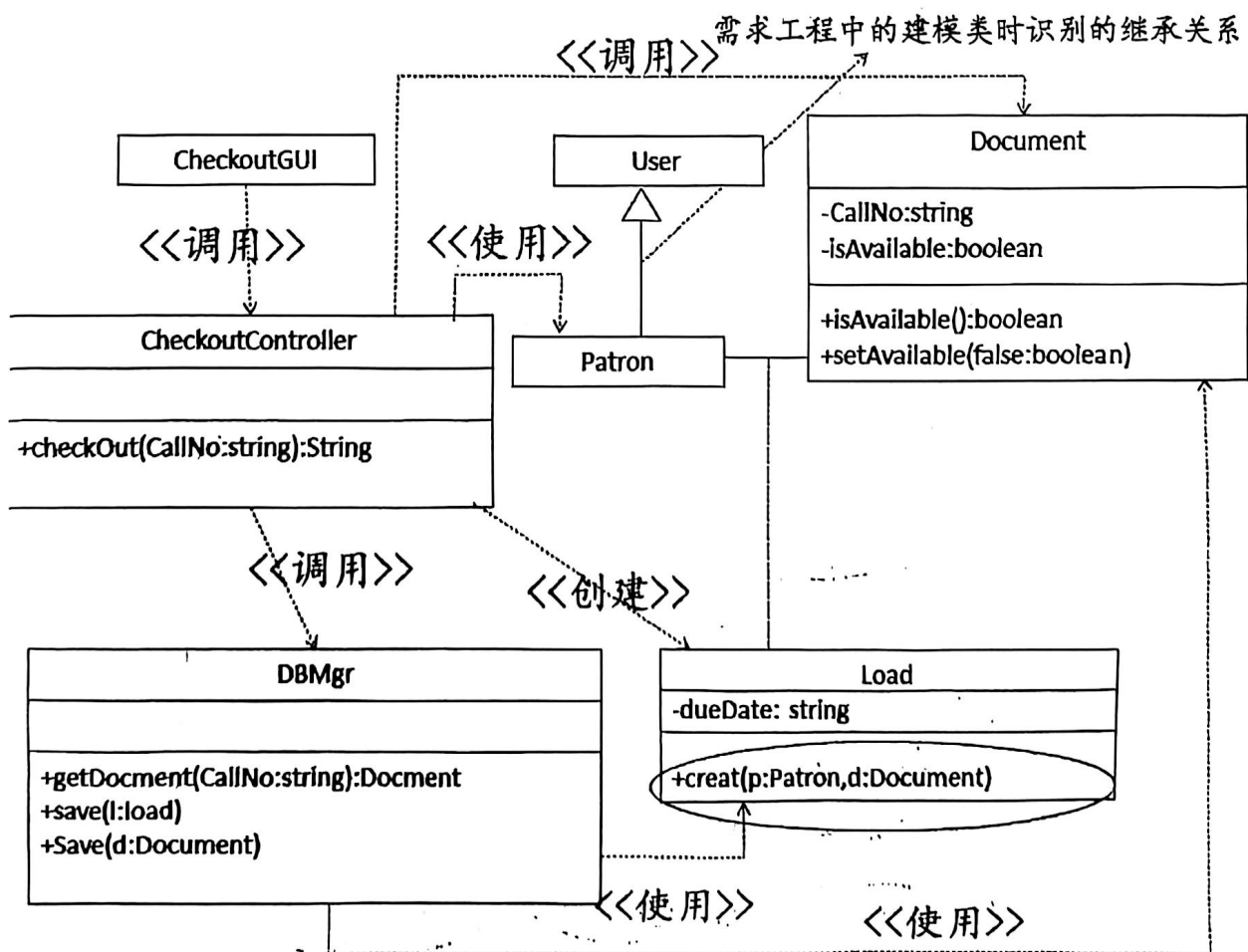
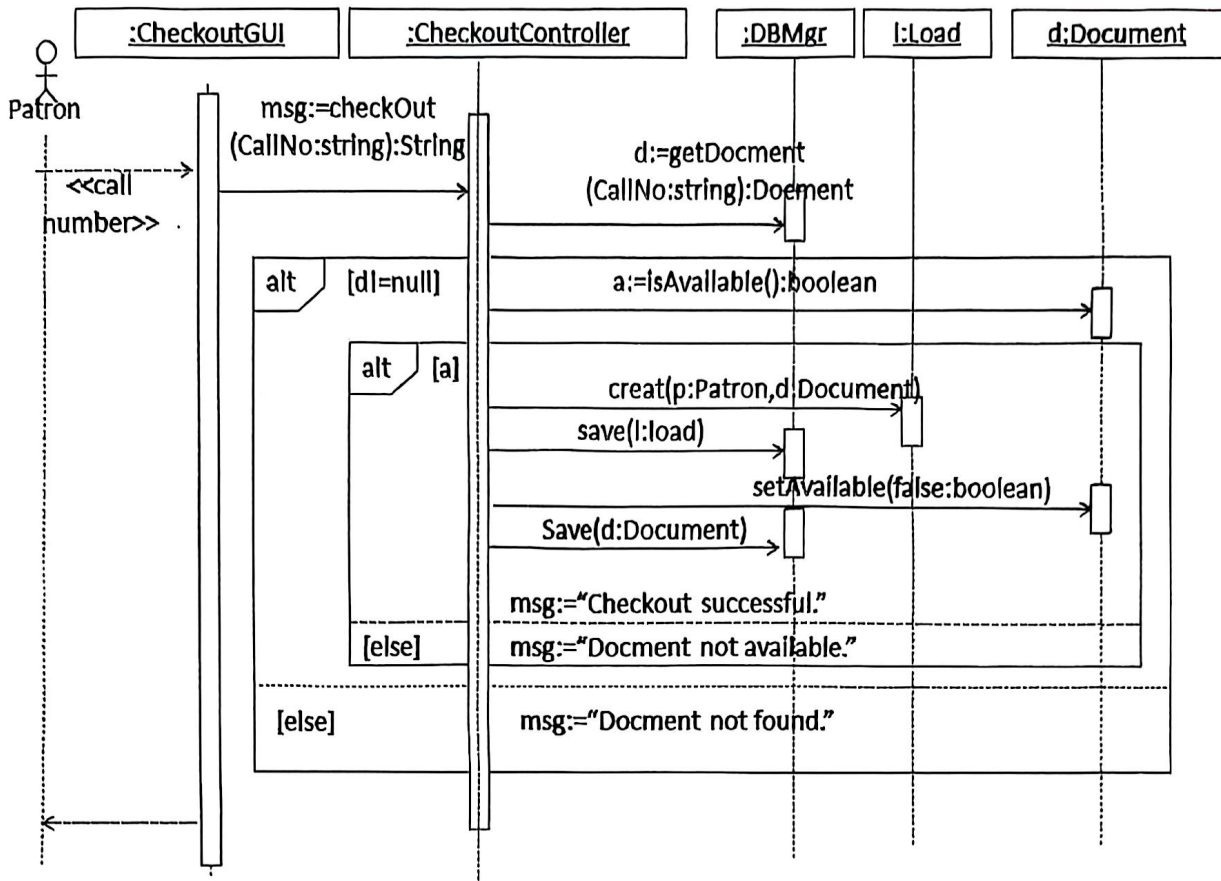
“爱因斯坦认为必然存在着一个对自然界简单的解释, 因为上帝既不专制也不喜怒无常。然而软件工程师却无法抱侥幸心理, 多数情况下, 他必须面对毫无规律的复杂性。”——Fred Brooks

第二题参考答案 (20分)

①



第三题参考答案 (15分)



第四题参考答案 (15分)

